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Title of the Experiment:

Salt and Pepper Separation Using Static Electricity

Background Information:

- **Describe** the validity of the research question be tested by scientific investigation
- **Explain** the reasoning behind the hypothesis using correct scientific reasoning

Validity of the Research Question: **If the balloon is statically charged by the fur, then there is a chance that the pepper could be more attracted to the balloon due to the fact that pepper is lighter. The purpose of this experiment is to find the limits of static electricity and to find out what it can achieve.**
Reasoning behind the Hypothesis: **If the balloon is charged by the fur, then the pepper will either be pushed away or drawn towards the balloon, because the balloon becomes statically charged and the lighter pepper pieces are easier to lift than the heavier salt grains.** **It works because pepper is lighter than salt, so while both salt and pepper are attracted to an object charged with static electricity, only the lighter pepper jumps up ("Separate Salt and Pepper with Static Electricity").**

Research Question:

- Must be based on the **description** given in the **Background Information**
- Must be in the form of a question

Can salt and pepper be separated using static electricity?

Hypothesis:

- Must be based on the reasoning given in the **Background Information**
- **State** the hypothesis in an "if, then, because," format
- Must **state** all variables used, followed by **one** chosen variable

If the balloon is charged by the fur, then the pepper will either be pushed away or drawn towards the balloon, because the balloon becomes statically charged and the lighter pepper pieces are easier to lift than the heavier salt grains. ("Separate Salt and Pepper with Static Electricity")

Variables:
- **Identify** the dependent variable and the independent variable
- **Describe** how the data for each variable will be collected

Variable	The variable is:	Method of Measurement
Dependent	The salt and pepper separating.	Whether or not the salt and pepper separate.
Independent	The amount of salt and pepper used.	Sight.

Control Variables:
- **Identify** each control variable **(add more rows to the table if needed; not all rows need to be used)**
- **Describe** the method used to allow each variable to be controlled

The Control Variable is:	Method of Control
The amount of time spent rubbing the fur on the balloon	Counting
The types of salt and pepper	Using the same type of salt and pepper
Using the same fur and balloon	By using the same fur and balloon

Materials:
- Should be in a list (not a paragraph); you can bullet the list

- Should be as specific as possible (ie. don't just say "beakers," say Ten 250 mL beakers, for example)

- **One balloon**
- **One piece of rabbit fur**
- **Salt**
- **Pepper**

Method:

- Should be written in the past tense
- Should be numbered - <u>do not</u> use bullet points
- Should be clear and concise - **repeat steps** should be used wherever possible
- Should be logical, safe, and complete

1. **The furry side of the rabbit fur was rubbed against the side of the inflated balloon**

2. **A decently large amount of salt and pepper was poured onto the table (Around a square inch)**

3. **The statically charged side of the balloon faced the salt and pepper**

4. **Step one was repeated**

5. **A small amount of salt and pepper was poured onto the table (Around a square centimeter)**

6. **Step three was repeated.**